

Module 1 Cheat Sheet: Python Basics

| Package/Method | Description                                                                                         | Code Example                                                                                                                                                            |
|----------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Comments       | Comments are lines of text that are ignored by the Python interpreter when executing the code<./td> | <pre># This is a comment</pre>                                                                                                                                          |
| Concatenation  | Combines (concatenates) strings.                                                                    | <div>Syntax:<pre>concatenated_string = string1 + string2</pre></div> <div>Example:<pre>result = "Hello" + " John"&lt;/td&gt;</pre></div>                                |
| Data Types     | - Integer - Float - Boolean - String                                                                | <div>Example:</div> <pre>x=7 # Integer Value y=12.4 # Float Value is_valid = True # Boolean Value is_valid = False # Boolean Value F_Name = "John" # String Value</pre> |
| Indexing       | Accesses character at a specific index.                                                             | <div>Example:</div> <pre>my_string="Hello" char = my_string[0]</pre>                                                                                                    |
| len()          | Returns the length of a string.                                                                     | <div>Syntax:</div> <pre>len(string_name)</pre>                                                                                                                          |

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|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>Example:</p> <pre>my_string="Hello" length = len(my_string)</pre>                                                                                                                                                                           |
| lower()          | Converts string to lowercase.                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>Example:</p> <pre>my_string="Hello" uppercase_text = my_string.lower()</pre>                                                                                                                                                                |
| print()          | Prints the message or variable inside `()`.                                                                                                                                                                                                                                                                                                                                                                                                     | <p>Example:</p> <pre>print("Hello, world") print(a+b)</pre>                                                                                                                                                                                    |
| Python Operators | <ul style="list-style-type: none"><li>- Addition (+): Adds two values together.</li><li>- Subtraction (-): Subtracts one value from another.</li><li>- Multiplication (*): Multiplies two values.</li><li>- Division (/): Divides one value by another, returns a float.</li><li>- Floor Division (//): Divides one value by another, returns the quotient as an integer.</li><li>- Modulo (%): Returns the remainder after division.</li></ul> | <p>Example:</p> <pre>x = 9 y = 4 result_add= x + y # Addition result_sub= x - y # Subtraction result_mul= x * y # Multiplication result_div= x / y # Division result_fdiv= x // y # Floor Division result_mod= x % y # Modulo&lt;/td&gt;</pre> |
| replace()        | Replaces substrings.                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>Example:</p> <pre>my_string="Hello" new_text = my_string.replace("Hello", "Hi")</pre>                                                                                                                                                       |
| Slicing          | Extracts a portion of the string.                                                                                                                                                                                                                                                                                                                                                                                                               | <p>Syntax:</p> <pre>substring = string_name[start:end]</pre> <p>Example:</p> <pre>my_string="Hello" substring = my_string[0:5]</pre>                                                                                                           |

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| split()             | Splits string into a list based on a delimiter. | Example:<br><pre>my_string="Hello"<br/>split_text = my_string.split(",")</pre>                                                                            |
| strip()             | Removes leading/trailing whitespace.            | Example:<br><pre>my_string="Hello"<br/>trimmed = my_string.strip()</pre>                                                                                  |
| upper()             | Converts string to uppercase.                   | Example:<br><pre>my_string="Hello"<br/>uppercase_text = my_string.upper()</pre>                                                                           |
| Variable Assignment | Assigns a value to a variable.                  | Syntax:<br><pre>variable_name = value</pre><br>Example:<br><pre>name="John" # assigning John to variable name<br/>x = 5 # assigning 5 to variable x</pre> |

**Skills Network**